

Erick C. Jones Jr., PhD, PE

ASSISTANT PROFESSOR · INDUSTRIAL, MANUFACTURING, AND SYSTEMS ENGINEERING

The University of Texas at Arlington, Woolf Hall 420X, 500 West First St., Arlington, TX 76019, United States

✉ erick.jones@uta.edu | 🏠 www.erickjonesphd.com

Education

University of Texas at Austin

PH.D. OPERATIONS RESEARCH AND INDUSTRIAL ENGINEERING

- Advisor: Benjamin Leibowicz
- Dissertation: Multi-Systems Optimization: Intermittent Production, Flexible Demand, Emerging Technologies
- Graduate Portfolios: Food-Energy-Water Systems | Energy Studies

Austin, TX

August 2021

University of Texas at Austin

M.S. OPERATIONS RESEARCH AND INDUSTRIAL ENGINEERING

Austin, TX

May 2019

Texas A&M University

B.S. CHEMICAL ENGINEERING; MINOR: PETROLEUM ENGINEERING

College Station, TX

May 2014

Academic & Research Appointments

- 2021- **Assistant Professor**, Industrial, Manufacturing, and Systems Engineering, University of Texas at Arlington
- 2024 **Research Fellow**, Faculty Advised Clean Energy Sciences (FACES), National Renewable Energy Laboratory
- 2017-2021 **Graduate Research Assistant**, Operations Research and Industrial Engineering, University of Texas at Austin
- 2020-2021 **Researcher**, Texas Energy Poverty Research Institute (TEPRI)
- 2020-2021 **Research Fellow**, The Mickey Leland Energy Fellowship (DOE), Los Alamos National Lab
- 2020 **GHG Model Developer**, CRANE, Prime Coalition
- 2009-2014 **Undergraduate Research Assistant**, Chemical Engineering, Texas A&M University

Teaching Experience

- 2026- **Energy Systems Modeling**, Assistant Professor, IMSE, University of Texas at Arlington
- 2023- **Operations Research I**, Assistant Professor, IMSE, University of Texas at Arlington
- 2023- **Advanced Operations Research**, Assistant Professor, IMSE, University of Texas at Arlington
- 2021- **Engineering Probability**, Assistant Professor, IMSE, University of Texas at Arlington
- 2018-2019 **Engineering Finance**, Teaching Assistant, ME, University of Texas at Austin
- 2018 **Experimental Fluid Mechanics**, Teaching Assistant, ME, University of Texas at Austin
- 2017 **Fluid Mechanics**, Teaching Assistant, ME, University of Texas at Austin
- 2018 **Planet Texas 2050**, Curriculum Developer, University of Texas at Austin

Industry Positions

- 2015-2017 **Project Engineer**, Johnson Controls *Houston, TX*
- 2014-2015 **Process Engineer**, AMEC Foster Wheeler *Houston, TX*
- 2012 **Engineering Intern**, HDL Power *Brenham, TX*
- 2011 **Engineering Intern**, PWD Group *Queretaro, MX*

PROFESSIONAL LICENSES

Professional Engineer License (161829)

Former Mortgage Loan Originator (NMLS ID 1383124; 2015-2017)

Awards and Fellowships

SPONSORED RESEARCH

2025-2030	RET Site: Artificial Intelligence and Machine Learning Approach for Sustainable and Resilient Infrastructure in Urban Communities , Co-PI, NSF, PI: Nur Yazdani, Co-PIs: Yichen Zhang, Shuo Linda Wang, Maricela Leon	\$599,885
2025-2030	An Integrated Framework for Next-Gen NWS Surveillance: Utilizing Smart Trap Technology, Targeted Lures and Public Reporting Networks , Co-PI, USDA (APHIS), PI: Shouyi Wang, Co-PIs: Prasanna Gowda, Jianzhong Su, Wei-Jen Lee, Yijing Xie	\$4,562,751
2026-2031	Land Based Testing of Shipboard Power Systems , Co-PI, Naval Sea Systems Command (NAVSEA), PI: David Wetz Jr., Co-PIs: Alexander Johnston, Ankur Jain	\$86,416,590
2025-2027	Technical Support of NAVSEA's DDG(X) and Long Operation Combatant Naval Energy Storage System (LOCNESS) Programs , Co-PI, Office of Naval Research, PI: David Wetz Jr., Co-PIs: Alexander Johnston, Ankur Jain	\$6,913,260
2024-2025	Development of a Discrete-Event Simulation Model for Multiple Queue, Multiple Contract Pavement Maintenance Queueing System , PI, City of Dallas: Department of Public Works	\$50,000
2023-2024	Green Warriors Weatherization Pilot with Southern Dallas Progress CDC and UTA , PI, City of Dallas Office of Environmental Quality & Sustainability	\$50,000
2023-2026	Engineering One Planet (EOP) Institutionalization Grant , PI, Lemelson and ASEE, CoPIs: Ann Cavallo, Andrew Clark, Kendra Wallis, Mengpi Zhan, Na-Li Kim	\$199,990
2023-2027	Preparing the Next-Generation Workforce to Design Advanced Electrical Power Systems for the DoD Deploying Multiple Sources, Energy Storage, Intelligent Control, and Transient Loads , Co-PI, DOD #81538-RT-REP, PI: David Wetz Jr.	\$793,659
2021-2025	RET: Sustainable and Resilient Infrastructure for Urban Communities , Co-PI, NSF #2055705, PI: Nur Yazdani, Co-PIs: Melanie Sattler, Victoria Chen, Muhammad Huda, Wei-Jen Lee, Kate Hyun	\$596,000
2022	Engineering One Planet (EOP) Mini-grant , PI, Lemelson and ASEE	\$8,000
2020-2023	EAGER: AI-Enabled Optimization of the COVID-19 Therapeutics Supply Chain to Support Community Public Health , PI (transferred), NSF #2028612	\$199,993
2020-2024	Understanding the Impact Graduate Fellowships on Broadening Participation in GEO , PI (transferred), NSF #2028343	\$66,488

FELLOWSHIPS

2024	Faculty-Advised Clean Energy Sciences (FACES) Program , National Renewable Energy Lab
2024	Energy and Climate Tech-Innovation Policy Bootcamp , Bipartisan Policy Institute; George Mason School of Policy and Government; University of Maryland School of Public Policy
2023-2025	Energy Policy Advisory Council , Powerhouse Texas
2022	Service Learning Fellow , UT Arlington
2022	Faculty Advisory Sustainability Network (FANS) Fellowship , UT Arlington
2019-2021	National Research Trainee: INFEWS Scholar Program , National Science Foundation
2020	Mickey Leland Energy Fellowship , Department of Energy
2017-2021	GEM Fellowship , The National GEM Consortium

AWARDS

2025	MIF Early Career Award Honorable Mention , INFORMS
2010-2014	Louis Stokes Alliance for Minority Participation Scholarship (LSAMP) , NSF
2012	Artie McFarren Department of Chemical Engineering Scholarship , Texas A&M University
2012	12th Man Kickoff Team Scholarship , Texas A&M University
2011	Distinguished Student Award for Outstanding Academic Achievement , TAMU
2010	Dwight Look Engineering College Scholarship , Texas A&M University

Publications

PEER-REVIEWED RESEARCH ARTICLES

- Jones, E.C., Jr.;** Jones, E.C., Sr. Megawatts to Zettaflops: A Techno-Economic Framework for Grid-Tied Behind-the-Meter Architectures in AI Data Centers. *Electricity* 2026, 7, 43. <https://doi.org/10.3390/electricity7020043>
- Akhter, R., Palli, S. Walanjuwani, M.; **Jones, E.C., Jr.** Mapping the Supply Chain of Lithium-Ion Battery Metals from Mine to Primary Processing by Country and Corporation. *Commodities* 2026, 5(1), 2. <https://doi.org/10.3390/commodities5010002>
- Patil, H., Kumbhar, A.; **Jones, E.C., Jr.** Contributions of Extended-Range Electric Vehicles (EREVs) to Electrified Miles, Emissions and Transportation Cost Reduction. *Energies* 2025, 18(24), 6448. <https://doi.org/10.3390/en18246448>
- Jones, E.C., Jr.** Climate and Trade Policy for Risk Management: The Need for Geopolitical Analytical Frameworks for Supply Chain and Energy System Modeling. *Current Sustainable/Renewable Energy Reports* 2025, 12(9). <https://doi.org/10.1007/s40518-025-00256-x>
- Jones, E.C., Jr.;** Sridharan, C.; Aghapour, R.; Rodriguez, A. Re-Energizing Legacy Fossil Infrastructure: Evaluating Geothermal Power in Tribal Lands and HUBZones. *Sustainability* 2025, 17(6), 2558. <https://doi.org/10.3390/su17062558>
- Atitebi, O.S.; Dumre, K.; **Jones, E.C., Jr.** Centralized vs. Decentralized Black-Mass Production: A Comparative Analysis of Lithium Reverse Logistics Supply Chain Networks. *Logistics* 2025, 9(3), 97. <https://doi.org/10.3390/logistics9030097>
- Atitebi, O.S.; Dumre, K.; **Jones, E.C., Jr.** Supporting a Lithium Circular Economy via Reverse Logistics: Improving the Preprocessing Stage of the Lithium-Ion Battery Recycling Supply Chain. *Energies* 2025, 18, 651. <https://doi.org/10.3390/en18030651>
- Reeves, J., Griner, S., Johnson, K., **Jones, E.C., Jr.**, Shangani, S., “Exploring Relationships Between Dating App Use and Sexual Activity among Young Adult College Students”. *Frontiers in Reproductive Health*, Vol. 6., November 2024. DOI:10.3389/frph.2024.1453423
- Jones, Erick C., Jr.** Lithium Supply Chain Optimization: A Global Analysis of Critical Minerals for Batteries. *Energies* 2024, 17, 2685. DOI:10.3390/en17112685
- Biney, J., **Jones Jr., E. C.** and Jones, E. C. “Understanding the EV Semiconductor Chip Sustainable Supply Chain Chip Shortage” *International Supply Chain Technology Journal*, vol. 10, no. 2, 2024, <https://doi.org/10.20545/isctj.v10i2.336>
- Erick C. Jones, Jr.**, and Ariadna Reyes. 2023. “Identifying Themes in Energy Poverty Research: Energy Justice Implications for Policy, Programs, and the Clean Energy Transition” *Energies* 16, no. 18: 6698. <https://doi.org/10.3390/en16186698>
- Varekar, S., Srinivasan, S., Shah, N. Gopalakrishnan, N. N., **Jones Jr., E. C.**, , “Analyzing the Connectivity of Combined Statistical Areas in Different Census Regions Using ArcGIS” *International Supply Chain Technology Journal*, vol. 9, no. 10, 2023, <https://doi.org/10.20545/isctj.v9i10.319>
- Jones, Erick C.**, Sean Yaw, Jeffrey A. Bennett, Jonathan D. Ogland-Hand, Cooper Strahan, and Richard S. Middleton. 2022. “Designing Multi-Phased CO2 Capture and Storage Infrastructure Deployments.” *Renewable and Sustainable Energy Transition* 2 (August): 100023. <https://doi.org/10.1016/J.RSET.2022.100023>.
- Erick C. Jones Jr.** and Benjamin Leibowicz , “Climate Risk Management in Agriculture Using Alternative Electricity and Water Resources: A Stochastic Programming Framework”, *Environment, Systems, and Decisions*, 2021, DOI:10.1007/s10669-021-09838-8
- Jones EC, Azeem G, **Jones EC**, Jefferson F, Henry M, Abolmaali S and Sparks J (2021) Understanding the Last Mile Transportation Concept Impacting Underserved Global Communities to Save Lives During COVID-19 Pandemic. *Front. Future Transp.* 2:732331. doi: 10.3389/ffutr.2021.732331
- Erick Jones** and Benjamin D. Leibowicz. “Co-Optimization and Community: Maximizing the Benefits of Distributed Electricity and Water Technologies.” *Sustainable Cities and Society* Vol. 64, January 2021. DOI: 10.1016/j.scs.2020.102515
- Jones, E. C., Azeem, Gohar, **Jones, Erick C.**, and Jefferson, F., “Impacting at Risk Communities using AI to optimize the COVID-19 Pandemic Therapeutics Supply Chain”, *International Supply Chain Technology Journal (ISCTJ)*, Vol. 6, No. 9 September 2020. DOI: 10.20545/isctj.v06. i09.02
- Erick Jones** and Benjamin Leibowicz “Contributions of Shared Autonomous Vehicles to Climate Change Mitigation”, *Transportation Research Part D: Transport and Environment*, v. 72, 2019, pg. 279-298, 1361-9209, DOI: 10.1016/j.trd.2019.05.005.
- Juan C. Burgos, **Erick Jones**, and Perla B. Balbuena “Dynamics of Topological Defects in Single-Walled Carbon Nanotubes during Catalytic Growth”, *The Journal of Physical Chemistry C* 2014 118 (9), 4808-4817 DOI: 10.1021/jp412708h

Juan C. Burgos, **Erick Jones**, and Perla B. Balbuena “Effect of the Metal–Substrate Interaction Strength on the Growth of Single-Walled Carbon Nanotubes”, *The Journal of Physical Chemistry C* 2011 115 (15), 7668-7675 DOI: 10.1021/jp200919j

PEER-REVIEWED CONFERENCE PAPERS

Chanuk Lee, Ariadna Reyes-Sanchez, Luis Perez-Cortes, Ramya Aroul, **Erick Jones**, June Young Park. 2026. NZHR: An EnergyPlus-Integrated Serious Game with AI Teammate for Residential Retrofit Decision-Making. In *Proceedings of the 2026 ACM Sustainability Week (ACM Sustainability Week '26)*. Association for Computing Machinery, New York, NY, USA, 595–596. <https://doi.org/10.1145/3765611.3815351>

Javanpour, S., & **Jones, E.C., Jr.** (2026). A design and analysis of computer experiments process for sustainable building design. *IISE Annual Conference Proceedings*, Accepted

Jones, E.C., Jr. (2026). 3R-BET: Remanufacturing, repurposing, and recycling battery and energy technologies. *IISE Annual Conference Proceedings*, Accepted

Abbasi, D., Ghasemloo, M., & **Jones, E.C., Jr.** (2026). Project portfolio selection under uncertainty: A deterministic and stochastic optimization approach. *IISE Annual Conference Proceedings*, Accepted

Conlin, J., Conlin, G., & **Jones, E.C., Jr.** (2026). Solving the problem domain with systems engineering with the Power & Energy Innovative Research (PEIR) building. *IISE Annual Conference Proceedings*, Accepted

Bastóla, A., & **Jones, E.C., Jr.** (2026). Grid search–optimized ensemble learning framework for enhanced network intrusion detection. *IISE Annual Conference Proceedings*, Accepted

Alavi, S., & **Jones, E.C., Jr.** (2026). Quantum computing to solve a facility location optimization problem. *IISE Annual Conference Proceedings*, Accepted

Atitebi, O.S., & **Jones, E.C., Jr.** (2026). Designing experiments for optimized recovery of critical minerals from lithium-ion battery black mass: Bridging sustainability and process efficiency. *IISE Annual Conference Proceedings*, Accepted

Alavi, S., & **Jones, E.C., Jr.** (2026). Modeling policy-driven competition in the multi-regional lithium supply chain. *IISE Annual Conference Proceedings*, Accepted

Aghapour, R., & **Jones, E.C., Jr.** (2026). What really drives our power system models? Identifying the assumptions that shape U.S. power futures. *IISE Annual Conference Proceedings*, Accepted

Yang, E., Atitebi, O.S., Shelor, C.P., & **Jones, E.C., Jr.** (2026). Upcycling graphite from spent Li-ion batteries into graphene–nylon 12 composites for powder bed fusion additive manufacturing. *IISE Annual Conference Proceedings*, Accepted

Aghapour, R., Alavi, S., Atitebi, O.S., & **Jones, E.C., Jr.** (2025). Texas power system evolution and sectoral energy demand: Contrasting sector-specific and multi-sector model outcomes. *IISE Annual Conference Proceedings*, 1–6. https://doi.org/10.21872/2025IISE_8849

Mohanraj, M.R., Ramalingam, N., Ravichandran, K.R., Parameswaran, N., & **Jones, E.C., Jr.** (2025). Hands-on approach towards industrial engineering concepts through interactive Lego workstation assembly. *IISE Annual Conference Proceedings*, 1–6. https://doi.org/10.21872/2025IISE_8849

Markute, S.S., Bastóla, A., & **Jones, E.C., Jr.** (2025). Measuring microgrid degradation: Collecting meteorological and microinverter data. *IISE Annual Conference Proceedings*, 1–6. https://doi.org/10.21872/2025IISE_5962

Hassan Abadi, M.A., Saraswat, K., Adhikari, S., More, S., Ghasempour, M., & **Jones, E.C., Jr.** (2025). Designing a sustainable multi-objective mixed-integer linear programming (MILP) model for shrimp supply chains. *IISE Annual Conference Proceedings*, 1–6. https://doi.org/10.21872/2025IISE_8842

Aghapour, R., Alavi, S., Atitebi, O., **Jones, E.C., Jr.** Achieving Sustainable Energy Transition: A System Analysis Approach to Energy Policy Implementation. *Proceedings of the 2025 IISE Annual Conference*.

Jones, E.C., Jr., Wallis, K.L., Zhan, M.M., Kim, N., Cavallo, A.M., Clark, A. Teaching Teachers to Integrate Sustainability into Engineering Education: Lessons Learned from UTA's EOP Institutionization Program. *Proceedings of the 2025 ASEE Annual Conference & Exposition, Montreal, Quebec, Canada*. 10.18260/1-2–57198. <https://peer.asee.org/57198>

Zagozda, K., Wheelock, R., Aghapour, R., Rahman Mohammadpour, S., Sood, J., **Jones, E.C., Jr.**, Chen, V.C.P. Bringing College Level Engineering Research Experiences into the K-12 Classroom. *Proceedings of the 2025 ASEE Gulf Southwest Annual Conference, Arlington, TX*. 10.18260/1-2–55031. <https://peer.asee.org/55031>

Aghapour, R., **Jones, E.C., Jr.**, Alavi, S. (2024). Green Building Design Surrogate Optimization: Exploring Off the Shelf Machine Learning and Mixed Integer Programming Integrations. *Proceedings of the 9th North American Conference on Industrial Engineering and Operations Management*. <https://doi.org/10.46254/NA09.20240139>

- Alavi S., Attebi, O.S., Evans, T., **Jones, E.C.** (2024). Multi-Source Biofuel Supply Chain Network Optimization. Proceedings of the 2024 IISE Annual Conference.
- Thomason, L., J. Hovey, J. Sigler, R. Smalls, S. Rahman Mohammadpour, V. Sharma, J. Sood, **E. C. Jones, Jr.**, J. M. Rosenberger, and V. C. P. Chen (2024). Training Teachers to Employ Design and Analysis of Computer Experiments for Research on Sustainable Building Design. In Proceedings of the 2024 American Society for Engineering Education (ASEE) Annual Conference and Exposition, Portland, OR, June.
- Erick Jones**, Victoria C.P. Chen, Vishnu Sharma, Suman Gudikandula, RaHSirearl Smalls, Jocelyn Sigler “Training Teachers to integrate a Sustainability Assessment for Green Building and Retrofitting Project into a K-12 Classroom”, ASEE Gulf Southwest Annual Conference 2023
- Tam Kemabonta, **Erick Jones**, Dana Harmon, and Jason Pittman “A New Approach to Developing Community Solar Projects for LMI Communities in ERCOT’s Competitive Electricity Markets”, 2021 IEEE Global Humanitarian Technology Conference
- Dana Harmon, **Erick Jones**, Emory Wolfe, and Jacquie Moss “Pathways for DERs to Reduce Energy Burdens in Harris County”, 2020 ACEEE Summer Study on Energy Efficiency in Buildings proceedings
- Erick Jones** “Decomposing Systems: Illustrating the Utility of Distributed Energy Resources with Decomposition Techniques”, Institute of Industrial and Systems Engineers Annual Conference and Expo 2020

PUBLISHED REPORTS

- Erick C. Jones Jr.**, The Missing Link to Secure Battery Minerals: Intermediate Processing, UMD-GMU Boot Camp Policy Brief, 2026
- Jeffers, R., Kushner, D., Ntakou, E., Paaso, A., Chalamala, B., Jourdier, B., Rahmatian, F., Fotuhu-Firuzabad, F., **Jones, E.C., Jr.**, Tapia, J., Abbas, A., Price, T., Enabling Climate Adaptation and Mitigation through Grid Modernization, TR-124, IEEE PES, 2024, October, <https://dx.doi.org/10.17023/02vn-4d23>
- Dana Harmon, Tam Kemabonta, Jacquie Moss, **Erick Jones**, and Andrew Robison, “When the Lone Star Froze Over - Winter Storm Uri and the lived experiences of Texas low income communities”, Texas Energy Poverty Research Institute, 2021. <https://www.txenergypoverty.org/2021/07/when-the-lone-star-froze-over-winter-storm-uri-report/>
- Erick Jones**, Jacquie Moss, and Tam Kemabonta, “Lived Experiences of Winter Storm Uri”, Texas Energy Poverty Research Institute, 2021, <https://www.txenergypoverty.org/2021/07/when-the-lone-star-froze-over-winter-storm-uri-report/>
- Erick Jones**, “Racial Disparities in Energy”, Smart Energy Consumer Collaborative, 2021. <https://smartenergycc.org/racial-disparities-among-lower-income-energy-consumers-white-paper/>
- Erick Jones**, Dana Harmon, and Tam Kemabonta, “Energy and Emissions Savings Potential of Renewable Thermal Technologies in Houston, Texas”, Renewable Thermal Alliance CBEY Yale, 2021. <https://cbey.yale.edu/research/energy-and-emissions-savings-potential-of-renewable-thermal-technologies-in-houston-texas>

WORKING ARTICLES, PAPERS, AND REPORTS

- Freese, Lyssa; Spandagos, Constantine; Andrews, Abigail; **Jones, E.C., Jr.**; Dowling, Jackie; Rising, James; Crozier, Constance; Mirkouei, Amin; Luo, Qian; Gong, Weila; Schenuit, Felix; Bednar, Dominic, Bridging Research and Policy to Achieve a Clean Energy Future, Joule, 2025, Under Review
- Aghapour, R.; **Jones, E.C., Jr.** Policy Drivers of Texas Energy System Outcomes: Insights from Design and Analysis of Computer Experiments with GCAM-USA. Energy Policy, 2026, Submitted
- Alavi, S.; **Jones, E.C., Jr.** A Multi-Decision Maker Approach for Decentralized Supply Chain Optimization in Critical Mineral Networks. Supply Chain Analytics, 2026, Under Review
- Aghapour, R.; **Jones, E.C., Jr.** Navigating Texas’s Renewable Energy Boom: Addressing Challenges with Transmission Expansion Solutions. Applied Energy, 2026, Working
- Atitebi, O.S.; **Jones, E.C., Jr.** Multi-Technique Material Characterization of Black Mass from Spent Lithium-Ion Batteries. Clean Technologies and Recycling, 2026, Working
- Aghapour, R.; **Jones, E.C., Jr.** The Future of Texas Energy Under Two Different Modeling Approaches: GCAM vs. REEDS, Computers and Operations Research, 2025, Working
- Alavi, S.; **Jones, E.C., Jr.** Lithium Supply Chain Competition: A Two-Player Game Analysis, Applied Energy, Policy and Regulation, 2025, Working

Supervising

PHD STUDENTS / POST-DOCTORATES

2025-	John Conlin , PhD Graduate Research Assistant	UTA
2023-	Oluwatosin Attebi , PhD Graduate Research Assistant	UTA
2023-	Sara Alavi , PhD Graduate Research Assistant	UTA
2022-	Raziye Aghapour , PhD Graduate Research Assistant	UTA
2021-22	Shannon Abolmaali , Post-Doctorate	UTA

UNDERGRADUATE STUDENTS

2024-2026	Abinash Bastola , Undergraduate Researcher	UTA
2024-2025	Siddharth Shivaji Murkute , Undergraduate Researcher	UTA
2023-2025	Nathan Miranda , Undergraduate Researcher	UTA
2023-2026	Guillermo Conlin , Undergraduate Researcher	UTA
2023-2024	Baseel Atallah , Undergraduate Researcher	UTA
2023-2024	Tara Chandrakasem , Undergraduate Researcher	UTA
2023-2024	Dylan Hoang , Undergraduate Researcher	UTA
2023-2024	Anne Nguyen , Undergraduate Researcher	UTA
2022-2023	Cecilia Romero , Undergraduate Researcher	UTA

MS STUDENTS

2024-2026	Hritik Vivek Patil , MS Scholarship Award Research Assistant	UTA
2024-2026	Sisira Reddy Palli , MS Research Assistant Author	UTA
2023-2025	Toryon Evans , MS GEM Fellow Research Assistant	UTA
2023-2025	Syed Ajmal Shakeeb , MS Research Assistant	UTA
2023-2025	Intesar Husain , MS Scholarship Award Research Assistant	UTA
2023-2025	Chandramouli Munjurpet Sridharan , MS Scholarship Award Research Assistant	UTA
2023-2025	Ahmed Saboor Syed , MS Scholarship Award Research Assistant	UTA
2023-2025	Saqib Raza Mohammed , MS Research Assistant Author	UTA
2022-2024	Sanjeev Joshi , MS Scholarship Award Research Assistant	UTA
2022-2024	Syed Mufid , MS Scholarship Award MS Research Assistant	UTA
2022-2024	Shraddha Varekar , MS Scholarship Award Research Assistant	UTA
2022-2024	Eshwara Prasad Sridhar , MS Scholarship Award Research Assistant	UTA
2021-2022	Suman Gudikandula , MS Scholarship Award Research Assistant	UTA
2022-	Other MS / OPT Students , 100+ Student / OPT Researchers and Counting	UTA

Talks and Presentations

INVITED TALKS / PANELS

- July 28, 2026. *A Multi-Region Network Optimization Framework for Critical Mineral Supply Chain Resilience*. **Invited Talk:** Lawrence Livermore National Laboratory
- October 23, 2025. *Tariffs, Climate and Trade Policy, and Supply Chains: A Lithium Supply Chain Optimization Case Study*. **Invited Talk:** East Texas A&M University
- September 23, 2025. *AI Coffee Hour for Engineering Faculty: Build Smarter Teaching Tools*. **Invited Talk:** UT Arlington
- April 7, 2025. *Tariffs, Climate and Trade Policy, and Supply Chains: A Lithium Supply Chain Optimization Case Study*. **Invited Talk:** North Carolina State University
- January 31, 2025. *The Future of Critical Mineral Supply Chains*. **Invited Talk:** Colorado School of Mines
- November 20, 2024. *Emissions and Resource Adequacy in ERCOT*. **Invited Panel Moderator:** Texas Energy Summit
- November 7, 2024. *Revenue Generating Grid-Interactive and Efficient Buildings (GEBs)*. **Invited Talk:** Symposium on Building Energy Research (SymBER)

November 6, 2024. *EVs and Lithium Supply Chains*. **Invited Talk:** Colorado School of Mines

September 6, 2024. *Energy Transition Supply Chains*. **Invited Talk:** University of Arkansas

April 4, 2024. *Stochastic Optimization Modeling of an Agricultural, Water, and Energy System Under Future Water Uncertainty*. **Invited Talk:** Prairie View A & M University

March 1, 2024. *The Role of Recycling in the Lithium Supply Chain*. **Invited Talk:** University of Oklahoma

December 14, 2023. *Interim Charge Strategy Session*. **Invited Panel:** Energy Policy Advisory Council

November 7, 2023. *Grid Resilience and Community Impact Planning – Approaches and Lessons*. **Invited Panel:** IEEE PES Latin America

June 15, 2023. *Finding your way in Macro-Energy Systems*. **Invited Panel:** International Energy Workshop

March 28, 2023. *Build Better for Climate Panel*. **Invited Panel:** North Texas Climate Change Symposium

May 3, 2022. *A Stochastic Programming Framework for Resource Allocation on a Farm Under Climate and Weather Uncertainty*. **Invited Lecture:** UTA Research Institute

March 25, 2022. *Climate Risk Management in Agriculture Using Alternative Electricity and Water Resources: A Stochastic Programming Framework*. **Invited Webinar:** UTA Virtual Brown Bag

December 16, 2021. *Weatherization Strategies for North Texas*. **Invited: Virtual Presentation** City of Dallas & Dallas Library

November 18, 2021. *A Stochastic Programming Framework for Resource Allocation on a Farm Under Climate and Weather Uncertainty*. **Invited Webinar:** IISE Energy Systems Division

May 13, 2021. *Energy and Water Efficient Building for Affordable Housing Developers*. **Invited Talk:** Energy Opportunities Coalition (TEPRI)

April 29, 2021. *Renewable Thermal Technologies*. **Invited Talk:** Yale Center for Business and the Environment: RTT

February 25, 2021. *Racial Disparities in Energy*. **Invited Talk:** SECC Research Brief: Racial Disparities in Energy

August 11, 2020. *Pathways for Reducing Energy Burdens in Harris County*. **Invited Talk:** City Efficiency Leadership Council North Texas Round-table

PRESENTATIONS

June 2026. *EOP Institutionalization 2nd Year*. **Oral Presentation** ASEE Annual Conference

May 2026. *3R-BET*. **Oral Presentation** IISE Annual Conference

October 2025. *Lithium Supply Chains*. **Oral Presentation** INFORMS Annual Meeting

June 2025. *EOP Institutionalization 1st Year*. **Oral Presentation** ASEE Annual Conference

October 2024. *Lithium Supply Chains*. **Oral Presentation** INFORMS Annual Meeting

May 2024. *Lithium Supply Chains*. **Oral Presentation** IISE Annual Conference

October 17, 2023. *Energy Transition Supply Chains*. **Oral Presentation** INFORMS Annual Meeting

May 22, 2023. *Sustainable and Equitable Allocation of Energy Resources: The Effect of Overlying on Certain Energy Supply Chains*. **Oral Presentation** IISE Annual Conference

October 17, 2023. *Energy Transition Supply Chains*. **Oral Presentation** INFORMS Annual Meeting

May 22, 2023. *Sustainable and Equitable Allocation of Energy Resources: The Effect of Overlying on Certain Energy Supply Chains*. **Oral Presentation** IISE Annual Conference

March 16, 2023. *Training Teachers to integrate a Sustainability Assessment for Green Building and Retrofitting Project into a K-12 Classroom*. **Oral Presentation** ASEE GSW Conference

2022. *Reproducible Research*. **Oral Presentation:** UTA Virtual Brown Bag, Arlington, TX

2022. *Climate Risk Management in Agriculture*. **Oral Presentation:** UTA IMSE Seminar, Arlington, TX

2022. *Energy Resources' Effect on Energy Poverty*. **Oral Presentation:** INFORMS Annual Meeting, Indianapolis, IN

2022. *Agricultural Stochastic Optimization Climate Uncertainty*. **Oral Presentation:** Annual IISE Conference, Seattle, WA

2021. *Agricultural Stochastic Optimization Climate Uncertainty*. **Oral Presentation:** Annual INFORMS Meeting, Anaheim, CA
2021. *Stochastic Optimization Modeling of an Agriculture, Water and Energy System under Future Water Uncertainty*. **Oral Presentation:** Annual IISE Conference
2020. *Co-Optimization and Community*. **Oral Presentation:** Annual IISE Conference
2020. SimCCS Gulf Coast Project. **Oral presentation:** Mickey Leland Energy Fellowship Program Technical Forum
2020. Co-Optimizing Distributed Energy, Water, and Transportation: An Optimization Based Decomposition Approach **Poster:** Energy Week 2020, Austin, TX
- 2019, 2020. Building a Sustainable Future: Co-Optimizing Community Energy and Water, Transportation and Power. **Poster:** Texas Energy Summit and Graduate and Industry Networking Event, Austin, TX
- 2018, 2019. The Role of SAVs in Climate Change Mitigation. **Poster:** GEM Conference and Graduate and Industry Networking Event, Los Angeles, CA and Austin, TX
2018. The Role of SAVs in Climate Change Mitigation. **Oral presentation:** GEM Conference 2018, Los Angeles, CA.

Teaching Experience

Energy Systems Modeling (REE 4301), University of Texas at Arlington

Arlington, TX

INSTRUCTOR: DR. ERICK C. JONES JR.

2026-

- Created Course and Open Education Resource Book for Use.
- Introduce students to the idea of energy systems big and small, how to model energy systems, and how to make decisions that improve them.
- Developed and evaluated exams, hands on activities, slides, projects, and HWs

Operations Research I (IE 3315), University of Texas at Arlington

Arlington, TX

INSTRUCTOR: DR. ERICK C. JONES JR.

2023-

- Instructed students on deterministic operations research concepts including: linear programming, integer programming, goal programming, nonlinear programming, and game theory and helped them apply those concepts to solving numerical problems.
- Created and Guided them through a practical project where they had to collect data, acquire skills in writing and programming (Python), and apply the course knowledge in a realistic way.
- Developed and evaluated exams, hands on activities, slides, projects, and HWs

Advanced Operations Research (IE 5301), University of Texas at Arlington

Arlington, TX

INSTRUCTOR: DR. ERICK C. JONES JR.

2023-

- Instructed students on deterministic and stochastic operations research concepts including: linear programming, integer programming, goal programming, nonlinear programming, decision trees, Markov chains, Monte Carlo simulation, and queuing and helped them apply those concepts to solving numerical problems.
- Created and Guided them through a practical project where they had to collect data, acquire skills in writing and programming (Python), and apply the course knowledge in a realistic way.
- Developed and evaluated exams, hands on activities, slides, projects, and HWs

Engineering Probability (IE 3301), University of Texas at Arlington

Arlington, TX

INSTRUCTOR: DR. ERICK C. JONES JR.

2021-

- Instructed students on basic probability theory and hypothesis testing, apply those concepts to solving numerical problems, and collect data and perform descriptive and inferential statistical analysis
- Created and Guided them through a practical project and helped them acquire skills in writing and programming (R)
- Developed and evaluated exams, hands on activities, slides, projects, and HWs

Engineering Finance (ME 353), University of Texas at Austin

Austin, TX

INSTRUCTOR: DR. ERHAN KUTANOGLU

2018-2019

- Created and/or repackaged homework questions, case studies, and exam questions for a class of 150+ students.
- Led 3 sessions of 30+ students each through different case studies exploring key class ideas.
- Held office hours for students. Tutored, helped with conceptual problems, and guided them through various problems.
- Graded students work following a rubric. Fielded questions about exam grades and corrected as needed.
- Proctored exams for students. Answered exam related questions and maintained order.

Experimental Fluid Mechanics (ME 130L), University of Texas at Austin*Austin, TX***INSTRUCTOR: DR. ADELA BEN-YAKAR***2018*

- Attended appropriate trainings for leading the lab, answering common questions and difficulties, and proper laboratory procedure.
- Guided 45+ students in sections of 15+ students through labs, enforced safety rules, and ensured each student received good experimental results
- Graded students work following a rubric. Fielded questions about corrections as needed.

Fluid Mechanics (ME 330), University of Texas at Austin*Austin, TX***INSTRUCTOR: DR. VAIBHAV BAHADUR***2017*

- Held office hours for the 80+ students in the class, tutored them, helped them with conceptual problems, and guided them through applied problems.
- Graded students work following a rubric. Fielded questions about exam grades and corrected as needed.
- Proctored exams for students. Answered exam related questions and maintained order.

Scientist in Residence, University of Texas at Austin*Austin, TX***SUPERVISOR: DR. JAY BANNER***2018-2019*

- Engaged High School Physics students in the excitement of scientific discovery.
- Helped teachers address TEKS through innovative learning activities that incorporated university research.
- Attended appropriate trainings and professional development to build presentation, teaching, and story telling skills

Planet Texas 2050, University of Texas at Austin*Austin, TX***SUPERVISOR: DR. JAY BANNER***2018*

- Developed lesson plans around the core areas for Planet Texas 2050: Urbanization, Energy, Water, Ecosystem Services
- Worked in a multi-disciplinary team and met weekly to collaborate and discuss ideas
- Published finalized lesson plans on the SciRes website: https://www.esi.utexas.edu/outreach/k12-resources/pt2050_lessonplans/

Employment History

Assistant Professor, University of Texas at Arlington*Arlington, TX**2021-*

- Research: Founded and direct the Sustainable and Efficient Allocation of Resources (SEAR) Lab, building a research program in end-to-end energy and critical mineral systems; secured over \$300K in external funding as PI and over \$100M as Co-PI across federal, municipal, and foundation sponsors.
- Research: Advise four doctoral students, one postdoctoral researcher, and dozens of master's and undergraduate researchers annually; published over 30 peer-reviewed articles and conference proceedings since 2021.
- Teaching: Serve as professor of record across five courses spanning the undergraduate core through the doctoral sequence, authoring all course material and assessing student mastery; built two new courses from scratch, including Energy Systems Modeling (REE 4301) and its accompanying open-access textbook.
- Service: Advise multiple student organizations; serve on university, departmental, and international professional committees; apply lab resources and findings to help K-12 organizations, municipal partners, and community non-profits achieve their goals.

Research Fellow, Faculty-Applied Clean Energy Sciences (FACES), National Renewable Energy Laboratory*Golden, CO**2024*

- Selected for the inaugural cohort of the Department of Energy's FACES program, hosted within NREL's buildings research group.
- Conducted cyberphysical experimentation on grid-interactive efficient buildings (GEBs).
- Developed a Customized Learning Solution translating the research into instructional material, delivered at UT Arlington and released for free public use.

Researcher, Texas Energy Poverty Research Institute (TEPRI)*Austin, TX**2020-2021*

- Conducted applied research on energy poverty, energy burden, and energy access among low-income Texas communities.
- Co-authored technical reports on the lived experiences of Winter Storm Uri and on renewable thermal technology potential, and contributed to analysis of racial disparities among lower-income energy consumers.
- Translated research findings into briefings and recommendations for utilities, regulators, and community organizations.

PhD Research Fellow, Mickey Leland Energy Fellowship (DOE), Los Alamos National Lab

Los Alamos, NM

2020-2021

- Explored carbon capture and storage deployment scenarios along the Gulf Coast using SimCCS.
- Applied optimization and operations research techniques to extend and improve the SimCCS modeling software.
- Contributed to a peer-reviewed publication on designing multi-phased CO₂ capture and storage infrastructure deployments.

GHG Model Developer, CRANE, Prime Coalition

Remote

2020

- Developed greenhouse gas emissions models for CRANE, an open-access tool for estimating the potential climate impact of early-stage climate technologies.
- Built and validated sector-level emissions accounting methods intended for use by impact investors and technology developers.

PhD Research and Teaching Assistant, University of Texas at Austin

Austin, TX

2017-2021

- Conducted doctoral research in multi-systems optimization, culminating in a dissertation on intermittent production, flexible demand, and emerging technologies.
- Supported my advisor's research portfolio, contributing to peer-reviewed work on shared autonomous vehicles, distributed electricity and water co-optimization, and climate risk management in agriculture.
- Served as teaching assistant in Mechanical Engineering for Engineering Finance (2018-2019), Experimental Fluid Mechanics (2018), and Fluid Mechanics (2017), leading recitations, proctoring, and grading assignments and exams.

Curriculum Developer, Planet Texas 2050, University of Texas at Austin

Austin, TX

2018

- Developed sustainability curriculum as a Scientist in Residence for Planet Texas 2050, the university's grand challenge research initiative.
- Translated university research on resilient water, energy, and urban systems into classroom-ready instructional material.

Project Engineer, Johnson Controls

Houston, TX

2015-2017

- Estimated, scheduled, forecasted, and managed secured HVAC equipment and controls projects ranging from \$50k to \$5MM.
- Created bids and managed the initiation of open projects across the same range, selling applied equipment into commercial and institutional facilities.
- Designed systems and processes for clients that satisfy their specifications, schedules, and code compliance.
- Oversaw job procurement, implementation, and overall execution on the construction site, and participated in start-up and delivery inspection of applied equipment.
- Executed a Six Sigma project identifying sources of variance and designed programs to reduce gross margin loss and forecasted revenue delays.
- Completed company training in project management, service and operations coordination, and the design, construction, and maintenance of duct, piping, and pump systems, compressors, air- and water-cooled chillers, and solution and custom air handlers.
- Together with process engineering practice at AMEC Foster Wheeler, this period constituted the qualifying engineering experience for Texas Professional Engineer licensure (P.E. No. 161829).

Process Engineer, AMEC Foster Wheeler

Houston, TX

2014-2015

- Created, modified, and backchecked process calculations, equipment datasheets, instrument datasheets, Process Flow Diagrams (PFDs), and Piping and Instrumentation Diagrams (P&IDs) with an emphasis on safety and efficiency.
- Calculated hydraulics for pressure safety valves (PSVs) and hydraulic circuits to verify proper line sizing, submitting calculation packages for client review on the LyondellBasell ethylene cracker in Corpus Christi, TX.
- Helped solve a coker antifoam problem by modernizing the design, rerouted light coker gas oil, and designed a new heat exchanger to improve system efficiency at ENAP Refinerias Bio Bio in Talcahuano, Chile.
- Checked fluid properties for pressure, level, and temperature instruments and verified pressure profiles at the Enterprise propane dehydrogenation plant in Mont Belvieu, TX.
- Designed or checked instruments, control valves, PSVs, pumps, and vessels across projects including the Exxon Mobil Antwerp refinery and a confidential paraxylene plant.
- Completed training in heat exchanger, piping and pump, pressure vessel, three-phase drum, coker, and turbine design and operation, and in chemical and oil and gas thermodynamics; recognized as joint Safety Member of the Month for Exxon BBSO / Foster Wheeler HEART observations.

Undergraduate Research Assistant, Texas A&M University*College Station, TX**2009-2014*

- Researched the growth mechanism of single-walled carbon nanotubes (SWCNTs) using specialized computer simulations under varying catalyst, substrate, temperature, and pressure conditions.
- Compared growth behavior with and without substrates and between pure metal and specialized alloy catalysts, organizing and interpreting simulation output with custom data analysis workbooks.
- Co-authored two peer-reviewed publications in The Journal of Physical Chemistry C (2011, 2014).
- Presented findings at technical conferences and research exhibitions, including a second-place technical exhibition award at the NSBE Region 5 conference.

Engineering Intern, HDL Power*Brenham, TX**2012*

- Used Six Sigma methodology to identify a number of problems with loss and efficiency across the manufacturing plant.
- Diagnosed problems in the manufacturing process for a newly introduced product.
- Created and implemented numerous solutions for the identified problems.
- Presented those solutions in several classes to employees and management to improve overall quality.

Engineering Intern, PWD Group*Queretaro, MX**2011*

- Used Design for Six Sigma methodology and techniques with an RFID focus to identify problems with loss and efficiency for consulting clients in food production, automotive components, and freight.
- Created a Spanish-language report and presentation documenting engineering observations and recommended improvements.
- Completed during a college semester abroad in Queretaro, Mexico.

Service, Outreach, & Professional Development

SERVICE AND OUTREACH (POST PHD)

2026-	IISE Energy Systems Division, Director	
2026-	Tellimus/Green+, Board Member	Dallas, TX
2026-	IMSE Undergraduate Curriculum Committee, Chair	Arlington, TX
2026-	College of Engineering Undergraduate Curriculum Committee, Committee Member	Arlington, TX
2026-	REE Undergraduate Curriculum Committee, Committee Member	Arlington, TX
2021-2026	Green Careers Texas, Board Member	Dallas, TX
2023-2025	Powerhouse Texas: Energy Policy Advisory Council, Advisory Council	Austin, TX
2026	IISE Annual Conference, Host Committee Member	Arlington, TX
2024-2025	Texas Energy Summit, Advisory Council	Austin, TX
2021-	TEPRI, Academic Advisory Committee Member	Austin, TX
2023-	NSBE Student Chapter at UTA, Faculty Advisor	Arlington, TX
2024-	INFORMS Student Chapter at UTA, Faculty Advisor	Arlington, TX
2025-2026	IMSE Undergraduate Curriculum Committee, Committee Member	Arlington, TX
2022-	IMSE Admissions Committee, Committee Member	Arlington, TX
2022-	IMSE Scholarship Committee, Committee Member	Arlington, TX
2024-2025	IEEE Task Force on Equity and Energy Justice, Member	
2022-2024	IEEE PES ITSCLC Climate Change Task Force, Member	
2024	INFORMS: Energy, Natural Resources, & the Environment, Energy Cluster Chair	Seattle, WA
2023	INFORMS: Energy, Natural Resources, & the Environment, Energy Cluster Chair	Phoenix, AZ
2021-	The GEM Consortium, University Representative	Arlington, TX
2023-2025	Fort Worth Regional Engineering and Science Fair, Judge	Arlington, TX
2023-2026	UTA Innovation Day, Faculty Advisor	Arlington, TX
2025	UTA REE Program Director Search Committee, Reviewer	Arlington, TX
2023	UTA EE Non-TT Professor Search Committee, Committee Member	Arlington, TX
2023	UTA EE Tenure-Track Professor Search Committee, Committee Member	Arlington, TX
2023	UTA Associate VPR Search Committee, Committee Member	Arlington, TX
2022	UTA Maverick Energy Institute Concept Group, Committee Member	Arlington, TX
2024-2026	Carter-Riverside Neighborhood Association, President	Fort Worth, TX
2024-2025	Fort Worth Transportation Plan: Moving 1 Million, Advisory Board	Fort Worth, TX
2023-2024	Carter-Riverside Neighborhood Association, Vice President	Fort Worth, TX
2022-2024	NSF Proposal Review, Reviewer	
2023	Energies Journal Special Edition on Energy Poverty, Guest Editor	
2023-	Energies Journal, Reviewer	
2024-	Sustainability Journal, Reviewer	
2024-	Discover Sustainability Journal, Reviewer	
2021-2023	Sustainable Cities and Society Journal, Reviewer	
2022	Urban Sustainability Journal, Reviewer	
2022	Energy Efficiency Journal, Reviewer	
2022	Annals of Operations Research Journal, Reviewer	
2021	Journal of Environmental Informatics, Reviewer	

CONFERENCES AND WORKSHOPS (2024 -)

6/2026	ASEE Annual Conference , Presenter	Charlotte
05/2026	IISE Annual Conference , Host Committee	Arlington, TX
11/2025	Texas Energy Summit , Advisory Board	Austin, TX
10/2025	INFORMS Annual Meeting , Presenter	Atlanta, GA
6/2025	ASEE Annual Conference , Presenter	Montreal
11/2024	Texas Energy Summit , Advisory Board	Austin, TX
11/2024	Symposium on Building Energy Research (SymBER) ,	Golden, CO
10/2024	INFORMS Annual Meeting , Energy, Natural Resources, and Environment Cluster Chair	Seattle, WA
06/2024	Energy and Climate Tech-Innovation Policy Bootcamp , Fellow	D.C.
05/2024	IISE Annual Conference , Presenter	Montreal

CONFERENCES AND WORKSHOPS (2021-2023, PRE-UTA THIRD YEAR REVIEW)

11/2023	IEEE PES Latin America, Panel Member	San Juan, PR
10/2023	INFORMS Annual Meeting, Energy, Natural Resources, and Environment Cluster Chair	Phoenix, AZ
07/2023	UTA Rising Junior Faculty DC Agency Visit, Rising Junior Faculty	D.C.
07/2023	IEEE PES General Meeting, Attendee	Orlando, FL
06/2023	International Energy Workshop, Panel Member	Golden, CO
05/2023	IISE Annual Conference, Presenter	New Orleans
03/2023	ASEE Gulf Southwest Conference, Presenter	Denton, TX
03/2023	North Texas Climate Change Symposium, Panel Member	UT Dallas
11/2022	Behavioral Energy and Climate Change Conference, Attendee	D.C.
11/2022	Regional Center of Expertise North Texas Annual Summit, Attendee	UT Dallas
10/2022	DOD HBCU/MI Opportunities Workshop, Attendee	Durham, NC
10/2022	INFORMS Annual Meeting, Energy Equity Flash Session	Indianapolis
10/2022	INFORMS Annual Meeting, New Faculty Colloquium	Indianapolis
10/2022	About Graduate School Workshop, Presenter	Arlington, TX
10/2022	"What is Research" Workshop, Presenter	Arlington, TX
2022-2023	Grants Academy, Attendee	Arlington, TX
2021-2023	Faculty Development Series for Research/Teaching/Service, Attendee	Arlington, TX
09/2022	NASA/TSU HBCU/MSI Technology Infusion Tour, Attendee	Houston, TX
09/2022	GEM 2022 Annual Board Meeting and Conference, Recruitment Table	Phoenix, AZ
07/2022	IEEE PES Annual Meeting, Climate Change Task Force	Denver, CO
06/2022	Faculty Advisor Network for Sustainability Conference, Attendee	Arlington, TX
05/2022	NSF ENG CAREER Workshop, Attendee	Online
05/2022	Annual IISE Conference, New Faculty Colloquium	Seattle, WA
05/2022	IEEE TD and Smart Grid Conference, Attendee	New Orleans
04/2022	Broadening Participation in STEM by Improved Financial Literacy Conference, Speaker	D.C.
04/2022	Understanding the Impact of Graduate Fellowships on Broadening Participation in Engineering and Geological Systems Conference, Speaker	D.C.
04/2022	RMI Whole Home Weatherization Summit, Attendee	Virtual
01-05/2022	Faculty Development Series for Research/Teaching/Service, Attendee	Arlington, TX
10/2021	Annual INFORMS Meeting, Session Chair: Agricultural Applications	Anaheim, CA
10/2021	NSF Virtual Grants Conference, Attendee	Virtual
09/2021	GEM 2021 Annual Board Meeting and Conference, Attendee	Houston, TX
08/2021	New Faculty Development Workshop, Attendee	Arlington, TX

SERVICE AND OUTREACH (PRE-PHD)

2019-2021	Institute for Operations Research and Management Sciences (INFORMS), Relations Chair	Austin, TX
2017-2020	Graduate Engineering Council, Vice President, New Activities Chair, Department Rep	Austin, TX
2020	City of Austin Sustainable Buildings Taskforce, Committee Member	Austin, TX
2019-2020	Urban Roots Farm, Project Development Volunteer	Austin, TX
2010-2014	National Society of Black Engineers (NSBE), Programs Chair, Officer	Texas A&M
2014-2017	United Way, Volunteer	Houston, TX
2016-2017	Baker Ripley, Volunteer	Houston, TX
2016	All-Earth Ecobot Challenge, Volunteer	Houston, TX
2014-2017	Big Brother Big Sisters, Volunteer	Houston, TX

CONFERENCES AND WORKSHOPS (PRE-PHD)

08/2021	GEM and NSF I-Corps Summer 2021 Entrepreneurship Program , Attendee	Virtual
2021	TEPRI's Energy Opportunities Coalition , Speaker / Attendee	Virtual
06/2021	National Academies of SEM Climate Conversations: Infrastructure , Attendee	Virtual
05/2021	IISE Annual Conference , Speaker	Virtual
05/2021	"Accelerating Sustainable Systems for Food Security in Extreme Environments and Food Deserts" NSF Convergence Accelerator Conference , Attendee	Virtual
04/2021	SimCCS Pro Workshop , Attendee	Virtual
04/2021	Energy Week 2021 , Attendee	Austin, TX
04/2021	BIPOC Climate Justice Dialogue , Attendee	Virtual
12/2020	The Energy Systems and Optimization Workshop , Attendee	Virtual
11/2020	INFORMS Annual Conference , Attendee	Virtual
11/2020	IISE Annual Conference , Speaker	Virtual
11/2020	UT COVID-19 Conference , Attendee	Virtual
10/2020	Inclusive Teaching and Learning Fall Symposium , Attendee	Virtual
09/2020	Stanford Workshop: Macro-Energy Systems , Attendee	Virtual
09/2020	The GEM Foundation Annual Conference , Attendee	Virtual
08/2020	Mickey Leland Energy Fellowship Program Technical Forum , Presenter	Virtual
08/2020	City Efficiency Leadership Council North Texas Roundtable , Presenter	Virtual
04/2020	Renewable Thermal Alliance Webinar , Presenter	Virtual
02/2020	Energy Week 2020 , Presenter	Austin, TX
02/2020	Graduate and Industry Networking Event , Volunteer	Austin, TX
2020	Sustainable Buildings Austin Climate Action Plan , Subject Matter Expert	Austin, TX
11/2019	Texas Energy Summit , Presenter	Austin, TX
06/2019	Alliance for Graduate Education and the Professoriate (AGEP) Research Exchange , Participant	Berkeley, CA
03/2019	3 Minute Thesis , Competitor	Austin, TX
02/2019	Graduate and Industry Networking Event , Volunteer	Austin, TX
09/2018	The GEM Foundation Annual Conference , Presenter	Los Angeles

PROFESSIONAL MEMBERSHIPS

Institute of Industrial and System Engineers (IISE)

Institute for Operations Research and the Management Sciences (INFORMS)

Macro-energy Systems (MES)

Institute of Electrical and Electronics Engineers (IEEE)

American Society for Engineering Education (ASEE)

The National GEM Consortium

National Society of Black Engineers (NSBE)